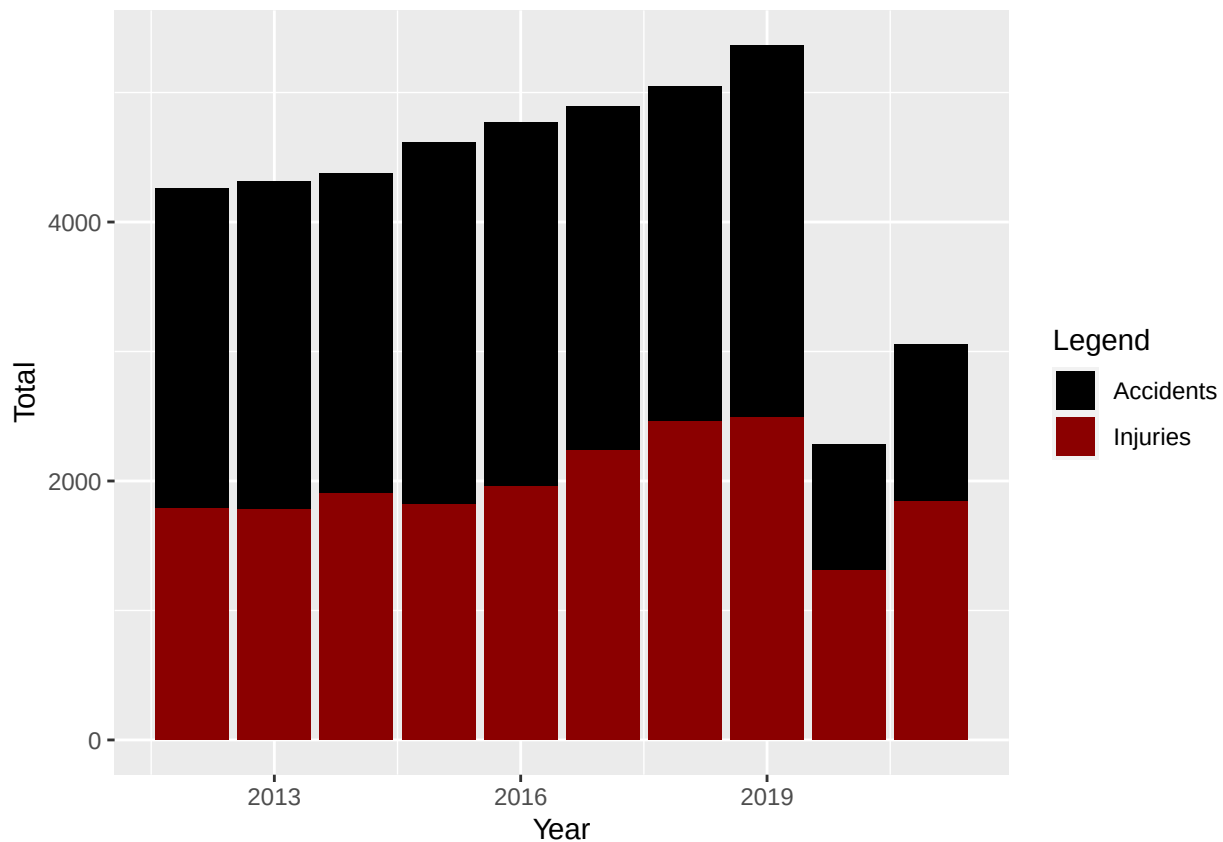


Assignment 5

Shanna Barrett

Tempe Crash Dataset

The Crash Data Report is a report of all crashes in Tempe, Arizona from 2012 to 2022. The dataset recorded things from latitude and longitude coordinates to age and gender of drivers, whether the driver was under the influence of drugs or alcohol, who was at fault and what the violation was, the severity of injuries, and other details about the crash. In the dataset, there were 42,969 entries.

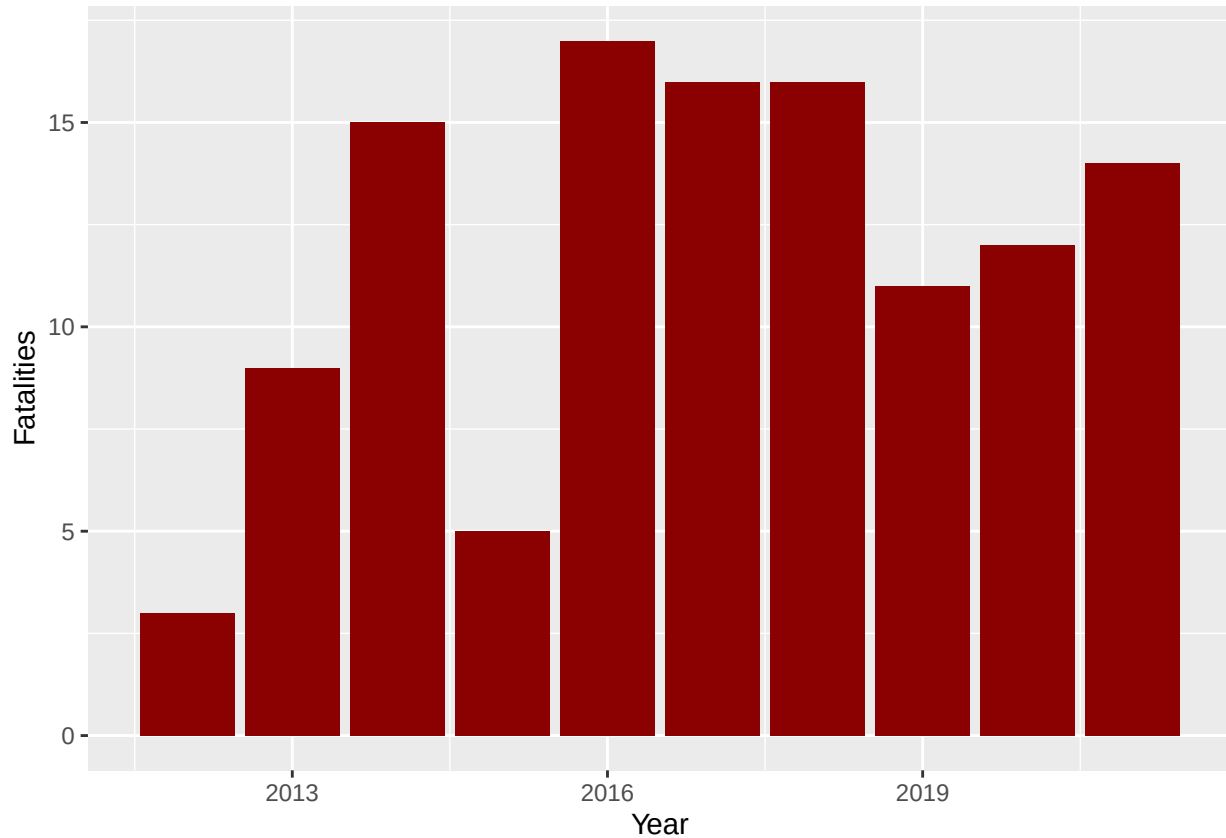


Accidents

This plot shows the total number of accidents and injuries from 2012-2021. With total number of accidents, in 2012-2014 there was a slow steady increase in accidents. There was a small jump from 2014-2015 and 2015-2018 there was a steady increase in accidents with another jump in 2019. Then there is a large decrease in the number of accidents in 2020. From 2020-2021, there is an about 700 increase in the number of accidents. The number of accidents in 2020 was about half as many as there was in 2019. The number of injuries roughly follows the trends of the accidents. One thing that differs is there was a decrease in the number of accidents in 2020 compared to 2014. The number of injuries compared to noninjuries is a larger proportion in accidents in 2020.

Fatalities

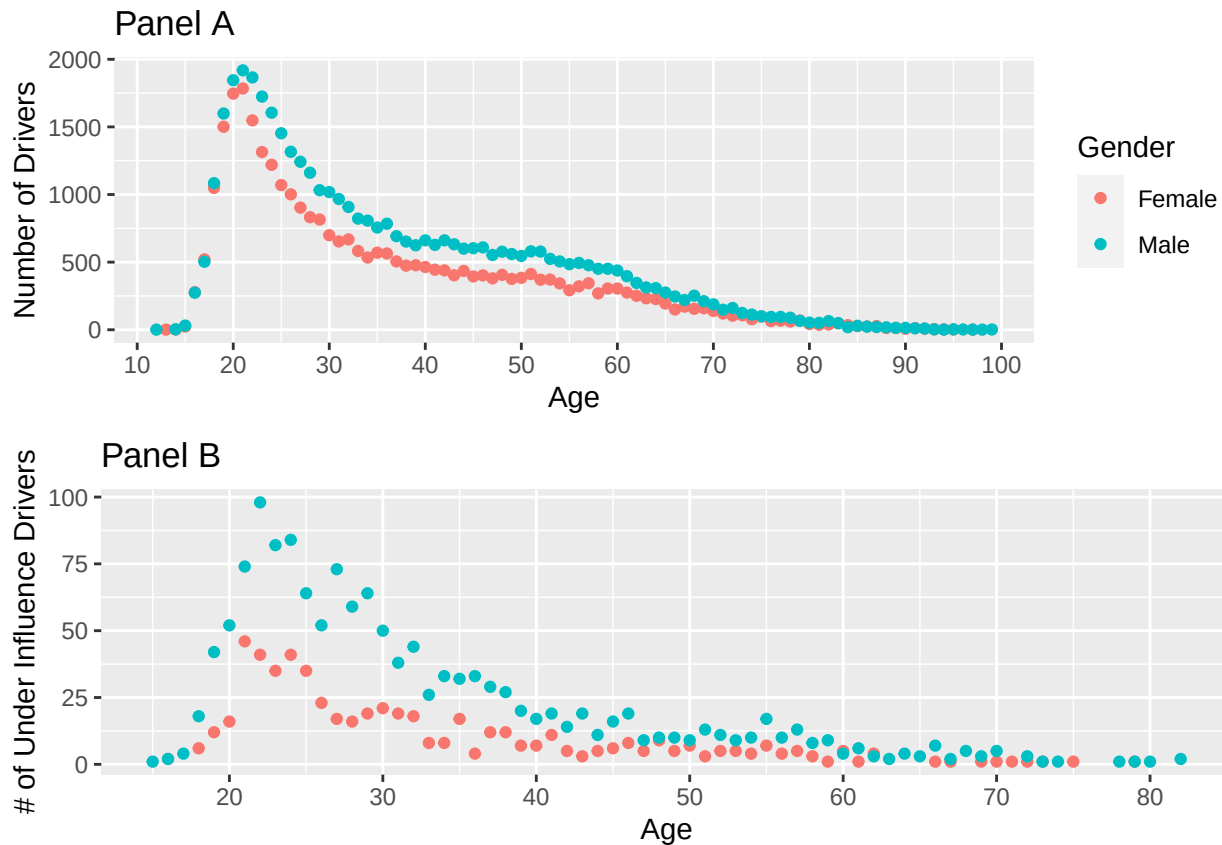
The number of fatalities has its own trend. There are large jumps from 2012 to 2013 and from 2013 to 2014. In 2015 the number decreases. The number jumps back up in 2016 and stays consistent for three years. The numbers decrease again in 2019 and have a small increase each year. The number of fatalities reflect the number of injuries in the first plot in how they both decreased from the previous year even though the number of incidents still increased.



The table below shows the number of people involved that were on drugs, alcohol, both, or none. To get this information, I had to split the crash data into individual one and individual two and then use rbind to stack the two tables and named the new table “Drivers”. This way I had all the information on the people who were involved in each accident. This then told how many people were involved depending on influence.

AlcoholUse	DrugUse	Total	Percent
Alcohol	Drugs	92	0.12
Alcohol	No Apparent Influence	1990	2.59
No Apparent Influence	Drugs	303	0.39
No Apparent Influence	No Apparent Influence	74437	96.90

The table reflects the number of people who fit into those categories. It does not reflect the number of accidents. Most of the people involved in an accident were not under any sort of influence. Only about 3% of people in the dataset were under one or more substance.



Panel A is showing the gender and number of drivers involved in an accident in Tempe from 2012-2021 who are the ages from 15 to 100. To create these two plots, I used the same “Drivers” table as the alcohol table. I filtered out the Genders that were unknown or null. I then filtered out the ages where the numbers weren’t plausible ages like “255” or anything above a hundred. The data was filtered where the Unit Type (Driver, Pedestrian, or Pedalcyclist) was only Drivers. Due to it being the age of drivers, not everyone involved in an accident, the ages start at 15 as that is the learners permit age in Arizona.

Panel A is showing the gender and number of drivers involved in an accident in Tempe from 2012-2021 who are the ages from 15 to 100. Females, from the ages, 19 to around 80 were less likely to be seen in an accident in Tempe. Panel B is showing the ages of drivers who were under the influence of alcohol while driving. The youngest male driver was 15 while the oldest male driver was 82. Females overall were involved in less accidents and were less likely to drive while under the influence of alcohol.

Conclusion

The Tempe Crash dataset was created to keep track of fatal accidents to start minimizing them in the future. Luckily, most of the accidents that happen in Tempe are not fatal, though a bit less than half do cause injury. Tempe has seen an increase in the number of accidents year after year until 2020 happened. That can be explained by stay-at-home orders due to COVID-19. With less people being on the road, the number of accidents dramatically decreased. Though the number of accidents decreased, the proportion of injuries to total accidents increased. The number of fatal accidents also didn’t decrease to the same degree as accidents. A person was more likely to be injured in a crash in 2020 than any other year. Males were also seen more often in accidents than were females. Age did play a part into how often you would see a specific gender. Ages 15 through 20, males and females were just as likely to be involved in an accident, regardless of substance use. After the age of 20, men were more often seen in accidents until the age of 80. After the age of 80, genders had similar numbers. When alcohol was involved, after the age of 18, males were recorded more often of being involved in a crash. The only time when females were recorded more often under the influence in a crash was ages 60, 62, and 75.

```

Crash_Data <- read.csv("~/Downloads/Crash_Data_Report_(detail).csv")

library(dplyr)
library(ggplot2)
library(reshape2)
library(lubridate)
library(tidyr)
library(kableExtra)
library(gridExtra)

#exploring the data
summary(Crash_Data)
unique(Crash_Data$Gender_Drv1)
unique(Crash_Data$Age_Drv1)
unique(Crash_Data$Lightcondition)
unique(Crash_Data$Violation1_Drv2)

Light_Conditions <- Crash_Data %>%
  group_by(Lightcondition) %>%
  filter(Lightcondition != "Unknown 51", Lightcondition != "51", )%>%
  tally()

Driver1 <- Personal_info %>%
  select(Year, Age_Drv1, Gender_Drv1, Unittype_One, Violation1_Drv1, AlcoholUse_Drv1, DrugUse_Drv1)

Driver2 <- Personal_info %>%
  select(Year, Age_Drv2, Gender_Drv2, Unittype_Two, Violation1_Drv2, AlcoholUse_Drv2, DrugUse_Drv2)

Driver1 <- Driver1 %>%
  rename(Age = Age_Drv1,
         Gender = Gender_Drv1,
         UnitType = Unittype_One,
         Violation = Violation1_Drv1,
         AlcoholUse = AlcoholUse_Drv1,
         DrugUse = DrugUse_Drv1)

Driver2 <- Driver2 %>%
  rename(Age = Age_Drv2,
         Gender = Gender_Drv2,
         UnitType = Unittype_Two,
         Violation = Violation1_Drv2,
         AlcoholUse = AlcoholUse_Drv2,
         DrugUse = DrugUse_Drv2)

Drivers = rbind(Driver1, Driver2)

#We are filtering out where Gender is unknown, and ages are not living ages.
#There is no note on what these numbers mean
#otherwise so we are filtering them out. They were all drivers as well so you
#can't assume it was driver 1 and an 11 year old.
#There wouldn't be that many 11 year old drivers
Drivers <- Drivers %>% filter(Gender != "", Gender != "Unknown", Age != 255, Age != 113, Age != 114,
                             Age != 115, Age != 116, Age != 117,
                             Age != 118, Age != 119, Age != 120, Age != 111, Age != 112, Age != 254)

```

```

Age_Drivers <- Drivers %>%
  group_by( Gender, Age) %>%
  filter(UnitType == "Driver")%>%
  tally()

A1 <- ggplot(Age_Drivers, aes(Age, n, col = Gender))+
  geom_point(stat = "identity")+
  xlim(0,100)+
  scale_x_continuous(breaks = seq(0, 100, by = 10))+
  labs(x = "Age", y = "Number of Drivers", title = "Panel A")+
  theme(legend.position = "right")

#This plot is showing the gender and number of drivers involved in an accident in Tempe from 2012-2021

influenced_driver <- Drivers %>%
  filter(AlcoholUse == "Alcohol")

influenced_driver <- influenced_driver %>%
  group_by( Gender, Age, AlcoholUse) %>%
  filter(UnitType == "Driver")%>%
  tally()

A2 <- ggplot(influenced_driver, aes(Age, n, col = Gender))+
  geom_point(stat = "identity")+
  xlim(0,100)+
  scale_x_continuous(breaks = seq(0, 100, by = 10))+
  labs(x = "Age", y = "# of Under Influence Drivers", title = "Panel B")+
  theme(legend.position = "none")

tally_alco <- Drivers %>%
  group_by(AlcoholUse, DrugUse)%>%
  tally()

tally_alco <- tally_alco %>%
  rename(Total = n)

tally_alco <- tally_alco %>%
  mutate(Percent = (Total/76822)*100)

tally_alco <- tally_alco %>%
  mutate(across('Percent', round, 2))

#This is the percentage of total people involved in accidents that
#were on drugs or alcohol or none
kable(tally_alco)

#1st to introduce the information, lets show total crashes and total injuries

Per_Year <- Crash_Data %>%
  group_by(Year) %>%
  tally()

```

```

Per_Year <- Per_Year %>%
  rename(TotalAccidents = n)

#ggplot(Per_Year, aes(x = Year, y = TotalAccidents))+
#  geom_bar(stat = "identity", fill = "darkred")+
#  labs(x = "Year", y = "Number of Accidents")+
#  ylim(0,6000)

Just_Injuries <- Crash_Data %>%
  filter(Totalinjuries != 0)

Just_Injuries <- Just_Injuries %>%
  group_by(Year)%>%
  summarize(Injuries = sum(Totalinjuries))

ggplot()+
  geom_bar(stat = "identity", data = Per_Year, aes(x = Year, y = TotalAccidents,
    fill = "Accidents"), size = 1)+
  geom_bar(stat = "identity", data = Just_Injuries,
    aes(Year, Injuries, fill = "Injuries"), size = 1)+
  labs(x = "Year", y = "Total", fill = "Legend")+
  scale_fill_manual(values = c("black", "darkred"))

#This is more accurate of all fatalities
Total_deaths <- Crash_Data %>%
  group_by(Totalfatalities)%>%
  summarize(Total_Deaths = sum(Totalfatalities))

Total_deaths <- Total_deaths %>%
  summarize(Total_Deaths = sum(Total_Deaths))

Just_Fatalities <- Crash_Data %>%
  filter(Totalfatalities != 0)

Just2_Fatalities <- Just_Fatalities %>%
  group_by(Year)%>%
  summarize(Fatalities = sum(Totalfatalities))

#Fatalities each year
ggplot(Just2_Fatalities, aes(x=Year, y= Fatalities))+
  geom_bar(stat = "identity", fill = "darkred")

grid.arrange(A1, A2)

```